Academic Statement of Purpose for Computer Science and Engineering

By Riley Paddock

The potential of Quantum Computers is exciting, with applications in chemistry, machine learning, optimization, and more. But honestly, in my time researching them I have become more interested in the problem of noise in quantum devices. The area is so rich with interesting approaches and methods from perturbative physics methods to mathematical error corrections. My research interests lie in approaching the presence of noise in quantum devices from two directions:

1. *How can we characterize and correct noise-induced errors in quantum systems?*
2. *How can noisy devices still be used to demonstrate advantage and solve problems?*

Answering both of these questions requires a careful balance of theory and practicality. A mixing of the theoretical models with the actual physics. I have worked closely with Professor Murphy Niu in both research and advanced coursework in developing more accurate and noise robust quantum controls and with Professor Prabhanjan Ananth on studying applications and verifications of NISQ devices.

RL-driven pulse design*:* Since January 2025 I have been working on Reinforcement Learning methods for designing quantum controls with Professor Murphy Niu. On most modern devices, the signals they send their qubits have a relatively simple structure. They ramp up the signal up to a certain amplitude, then just hold that amplitude for the right amount of time before ramping down[2]. This works well, but sacrifices a lot of the dimensionality of control. In reality, there are often signals which are both shorter, more accurate, and more robust to errors than any of these simplified ones. It has been shown that Machine Learning, specifically **Reinforcement Learning**, can navigate this space and it enables us **to find improved signals**[3]. Since control errors are a major contributor to noise in quantum computers, mitigating the control noise by choosing an optimal signal is very important to suppressing noise. My work in particular has been in finding higher quality multi qubit gates for a specific superconducting qubit architecture at the Advanced Quantum Testbed at Berkley Lab. This project was granted GPU hours through NVIDIA academic grants which enabled me to design and train powerful implementations of Proximal Policy Optimization (**PPO**) and Covariance Matrix Adapted Evolutionary Strategies (**CMA-ES**) for this approach. This involved carefully crafting a method to translate unrestrained model outputs into ansatz-free physical signals which then needed to be put through precise simulations of the device physics. Once this environment was designed to give our model maximal freedom and efficient evaluation, I designed the reward so the model would; produce physically feasible results, have freedom in gate time and amplitude, and was evaluated with some realistic stochastic noise. For example, I noticed that the model was outputting results very far outside the physical limits of our device which were getting rounded down to the physical maximum in the evaluation. So I added a penalty that slowly increased cost in the unphysical regions of the cost landscape. This way the model could be informed about the physical limits and be able to ‘surf’ the soft boundary to physical regions rather than being forced to bounce off complex boundary conditions. My work on PPO enabled us to find >0.99 fidelity three qubit gates for specialized superconductng qubit architecture[2]. My work with CMA-ES was able to find high-fidelity two qubit square root iSWAP and iSWAP gates that were better in both fidelity and feasibility than was we ever saw in fine-grained sweeps of common ansatz and runs of PPO. This improvement in the quality of two-qubit gates enables every circuit constructed of them to run faster and more accurately. This work is still in progress and preparation but we hope to share our findings by the end of this school year.

Refined theoretical models for error-robust pulses*:* I also worked on a novel technique in single qubit gate design in collaboration with Professor Murphy Niu and Professor Edwin Barnes at Virginia Tech. This technique, called **Space Curve Quantum Control,** maps quantum evolutions to curves in 3d space such that complex measures of error robustness (terms in the perturbative expansion) to simple geometric

properties of the 3d curve (Ex. Curve closure, Curvature, Torsion). It also allows you to restrict your pulse design choices to only those of high-fidelity and robustness to drift errors. I helped advance this approach by improving the machine feasibility of signals generated by these curves. My first approach was to add a fourier transform based feasibility constraint to the optimization workflow created by Virginia Tech, however this created a tradeoff with the other goals of the optimizer focused on error mitigation. So I developed a new description of the curves so that they were only built out of low-frequency components, then optimization could solely focus on error mitigation. This new description differed fundamentally from the previous structure so I had to verify that all the old geometric properties that were necessary for the formalism (could we specify the directions of the tangent and normal vectors at the endpoints, could we close the curve, etc) could still be satisfied. They all held and we are currently refining the method to integrate with the software made by Virginia Tech and writing a manuscript on the new approach. Overall, our approach brings the powerful theory of Space Curve pulse design closer to the reality of being implemented on-device.

Certified Randomness and Quantum Verification: Quantum verification is a protocol where a classical user can be guaranteed that they are connected to a quantum device, which is useful for Quantum Cloud applications such as certified randomness. With Professor Prabhanjan Ananth I investigated multiple approaches to develop such a protocol including Instantaneous Quantum Polynomial Circuits. My work was to review five papers on the technique with conflicting views on its security. I composed this into a special report breaking down; how the technique was developed, the security assumptions it proved, the methods of attack, the refinement to prevent attack, and a final analysis on its security. I also implemented the proposed attacks to test their effectiveness. This work resulted in one of the original authors, Dominik Hanglieter, visiting UCSB to share their perspective on the current state of IQP circuits.

In the Future: The problems I have spent time working on in quantum computing over the past year have endlessly fascinated me with their intricate depth and balance of real-world physical constraints and rich theoretical tools. The unique fusion of theory and practicality especially in a field so rich with active research and rapid development motivates me to continue to pursue techniques to mitigate error through creative co-design of architecture, control, and protocol. Recently, Google showed that RL can push devices running Quantum Error Correcting codes into lower regimes of logical error rates when paired with physics-informed device calibration[5]. I want to build and refine techniques just like that which require physics-informed calibration, cutting edge computer science methods, and intelligent method design to further reduce the influence of error in emerging quantum devices.

At Michigan: Michigan is becoming an increasingly important center for quantum technologies that I would be honored to contribute to. I have spoken to professor Gokul Ravi about his work in developing variational methods as well as designing robust error correcting code implementations and would love to contribute my experience with quantum control and machine learning to his design philosophy. Additionally, I would be excited to work with Professor Yunong Shi in developing compilation methods as well as implementing quantum error correction on unique architectures. Also, Michigan understands the inherent interdisciplinary collaboration important to quantum codesign with it's "Quantum Research Institute" and I would love to collaborate with engineers in the ECE department such as Yulong Dong in optimizing and designing quantum algorithms and Professor Sandeep Pradhan on projects in quantum information related to my experience with quantum verification. The work of these faculty aligns very closely with my interests and compliments very well with my experience and I am sure that a PhD at Michigan would enable me to make important contributions to the field and prepare me for work as a research scientist or an academic.

1. J. Zeng and E. Barnes, Fastest pulses that implement dynamically corrected single-qubit phase gates, Phys. Rev. A 98, 012301 (2018).

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4. H. Nelson, E. Piliouras, ET AL. Designing dynamically corrected gates robust to multiple noise sources using geometric space curves. Phys. Rev. A 108, 012407 (2023)
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